

# ISyE 6525 High Dimensional Data Analytics - HW3

## Module 3: Tensor Data Analysis

Fall 2026

**Submission.** Submit *one PDF report per question*, and for every question that requires code, *one code file for that question*. You may use MATLAB, R or Python. Total: **100 points**.

### Question 1.

(20 points)

#### Multilinear algebra.

(a) **(8 pts)** Let  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ . Compute by hand

$$(i) A \otimes B, \quad (ii) A \odot B, \quad (iii) A * B, \quad (iv) (A \otimes B)^{-1}, \quad (v) A^{-1} \otimes B^{-1},$$

where  $\otimes$ ,  $\odot$  and  $*$  denote the Kronecker, Khatri-Rao and Hadamard products. Comment on (iv) versus (v).

(b) **(6 pts)** Let  $A \in \mathbb{R}^{m \times n}$ ,  $B \in \mathbb{R}^{r \times s}$ ,  $C \in \mathbb{R}^{n \times p}$  and  $D \in \mathbb{R}^{s \times t}$ . Show that

$$(A \otimes B)(C \otimes D) = AC \otimes BD \in \mathbb{R}^{mr \times pt}.$$

(c) **(6 pts)** Let  $A \in \mathbb{R}^{m \times n}$ ,  $B \in \mathbb{R}^{p \times q}$ ,  $X \in \mathbb{R}^{q \times n}$  and  $Y \in \mathbb{R}^{p \times m}$ . Show that

$$(\text{vec}(Y))^{\top} (A \otimes B) \text{vec}(X) = \text{trace}(A^{\top} Y^{\top} B X),$$

where  $\text{vec}(\cdot)$  stacks the columns of a matrix on top of each other. *Hint:*  $\text{vec}(AXB) = (B^{\top} \otimes A) \text{vec}(X)$  and  $\text{vec}(U)^{\top} \text{vec}(V) = \text{trace}(U^{\top} V)$ .

### Question 2.

(20 points)

#### Rank of a tensor, and CP versus Tucker.

(a) **(12 pts)** Determining the rank of a tensor is NP-hard, but some upper bounds are useful in practice. Show that for  $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$

$$\text{rank}(\mathcal{X}) \leq \min\{IJ, JK, IK\}.$$

*Hint:* first show  $\text{rank}(\mathcal{X}) \leq IJ$  (or  $JK$ , or  $IK$ ) by writing the tensor as an explicit sum of rank-one terms.

(b) **(8 pts)** Show that the CP decomposition is a special case of the Tucker decomposition. Specifically, given  $\mathcal{X} = \sum_{r=1}^R \lambda_r a_r \circ b_r \circ c_r$ , exhibit a core tensor  $\mathcal{G}$  and factor matrices  $A, B, C$  such that  $\mathcal{X} = \mathcal{G} \times_1 A \times_2 B \times_3 C$ , and state how many free parameters each model has for a tensor of size  $I \times J \times K$ .

**Question 3.****(30 points)**

**Heat transfer process.** Consider a heat transfer process governed by

$$\frac{\partial S(x, y, t)}{\partial t} = \alpha \left( \frac{\partial^2 S}{\partial x^2} + \frac{\partial^2 S}{\partial y^2} \right),$$

where  $0 \leq x, y \leq 0.05$  is the location of an image pixel,  $\alpha$  is the thermal diffusivity coefficient and  $t$  is the time frame. The initial and boundary conditions are  $S|_{t=0} = 0$  and  $S|_{x=0} = S|_{x=0.05} = S|_{y=0} = S|_{y=0.05} = 1$ . At each time  $t$  the image is recorded on an  $n \times n$  grid of interior points; here  $n = 21$  and  $t = 1, \dots, 10$ , which gives ten  $21 \times 21$  images that are stored as a  $21 \times 21 \times 10$  tensor.

The thermal diffusivity coefficient depends on the material being heated. The file `heatT.mat` contains `tensor1` for material 1, `tensor2` for material 2, and a third tensor `tensor3` recorded from an unlabelled specimen.

- (a) **(12 pts)** Implement the CP-ALS algorithm from the slides yourself. Do **not** call `cp_als`, `tensorly` or any equivalent library routine. For each of the three tensors, run it for  $R = 1, \dots, 8$  and select the rank that minimizes

$$\text{AIC}(R) = 2 \left\| \mathcal{X} - \sum_{r=1}^R \lambda_r a_r \circ b_r \circ c_r \right\|_F^2 + 2R.$$

Plot AIC against  $R$  for the three tensors and report the optimal rank of each.

- (b) **(8 pts)** At the selected rank, plot the temporal loadings  $c_r$  of tensor 1 and display its leading spatial patterns  $\lambda_r a_r \circ b_r$  as images. Interpret them physically.
- (c) **(10 pts)** Determine which of the two materials `tensor3` corresponds to. Base your answer on the CP factors (state clearly what quantity you compare and why) and present the steps that lead to your conclusion.

**Question 4.****(30 points)**

**Data compression with Tucker truncation and hard thresholding.** Coefficient reduction is a key element of many data compression algorithms: after computing a transform of the input data, the least significant coefficients are discarded in order to decrease the memory footprint. Here we use the Tucker decomposition for dimensionality reduction and compact multidimensional data representation.

`Question4.mat` contains a third-order tensor  $\mathcal{X} \in \mathbb{R}^{64 \times 64 \times 40}$  holding 40 consecutive infrared inspection frames of a composite panel.

- (a) **(12 pts)** Write your own Tucker decomposition (do **not** call `tucker_als` or an equivalent)  $\mathcal{X} \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}$  using the ALS algorithm from the slides (initialize with HOSVD). Run it with  $R_1 = R_2 = R$  and  $R_3 = \min(R, 40)$  for  $R \in \{2, 4, 6, 8, 10, 15, 20, 25, 30, 40\}$  and report, for each  $R$ , the relative error

$$\varepsilon = \frac{\|\mathcal{X} - \hat{\mathcal{X}}\|_F}{\|\mathcal{X}\|_F}$$

and the compression ratio (number of entries of  $\mathcal{X}$  divided by the number of entries that must be stored, i.e. the core plus the three factor matrices). Plot relative error against  $R$  and against the compression ratio.

- (b) **(6 pts)** Show that when the factor matrices are column-wise orthonormal,

$$\varepsilon = \frac{\sqrt{\|\mathcal{X}\|_F^2 - \|\mathcal{G}\|_F^2}}{\|\mathcal{X}\|_F},$$

and verify the identity numerically on your  $R_1 = R_2 = 30$ ,  $R_3 = 20$  decomposition.

- (c) **(12 pts)** Fix  $R_1 = R_2 = 30$ ,  $R_3 = 20$ . Perform hard thresholding on the core tensor: keep only the 10%, 20%, 30%, 40% and 50% of core coefficients with the largest absolute value and set the rest to zero. Report the relative error in each case and display frames 5, 15 and 30 of the reconstruction. Comment on what this says about how the energy is distributed in the core.